

Dadmehr Daniel Ghasemfar

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SUMMARY

- Mechanical Engineer [B.S., Duke University] specializing in robotics, GPU physics simulations, and advanced prototyping, with minors in Mathematics and Psychology.
- Combines rigorous training in numerical methods, control systems, and fluid dynamics with hands-on expertise in CAD (SolidWorks, Ansys), additive manufacturing (SLS/metal 3D printing), and electronics.
- Co-founder of Atar Energy Solutions, designing disruptive high-voltage diagnostics tools at $\frac{1}{3}$ the cost and weight of competitors in a \$700M market. Led IP strategy and investor negotiations.
- Developed GPU based wave propagation simulation to train AI for acoustic perception, creating novel Chladni plate validation methods that bridged simulation and experimental data. Earned Duke's Graduation with Distinction award for this 2-year project.

TECHNICAL SKILLS

Languages: Java, Python, C++, MATLAB, Maple, TensorFlow, HTML/CSS/JavaScript, ROS

Software: CAD [SolidWorks, Fusion, OnShape], FEA and CFD [Ansys, SolidWorks, In-House], PCB [Eagle, KiCad]

Equipment: 3D Printing [SDF, SLS, Metal], Machining [Mill, CNC, Lathe, Woodwork], Lasercutting, Soldering

Theoretical: solving PDE's and ODE's, Numerical Analysis, Accurate GPU Simulations, PID's and Advanced Controls

Soft Skills: 6-month Project Management, Clear Communication, Professional Documentation, Fluent in Farsi, Conversational in Spanish

EDUCATION

Duke University

Durham, NC

Bachelor of Science in Mechanical Engineering with Minors in Math and Psychology

Aug 2020 – Dec 2024

- Total GPA: 3.48 → Engineering GPA: 3.56 → Math GPA: 3.5 → Psych. GPA: 3.0
- Classes: intro/intermediate robotics, intermediate dynamics, chaos and perturbation theory, control theory, heat and mass transfer, incompressible fluid dynamics, algorithms and data structures, ODE/PDE's, numerical analysis, intermediate solids and statics, aircraft performance and design
- Graduated as a Pratt Fellow for recognition of a 2+ year research project while maintaining strong grades.

Los Altos High School

Los Altos, CA

High School Diploma

Aug 2016 – June 2020

- Unweighted GPA: 3.86 → Weighted GPA: 4.35 → Credits: 320
- Classes: Multivariable Calculus, Linear Algebra, AP Calculus BC, AP Computer Science, Robotics II, AP Physics I/II/C, Advanced Scientific Investigations (Application based with two years of funded research), Intro to Engineering, AP Government and Economics.

- Graduated with recognition for my two years of research in Dr. Darren Dressen's advanced investigations lab. Awarded a Mountain View mayor's medal for my research on AI detected forest fires in 2019.
- Participated in the California Cross Country Championships race three years in a row, two of those years as a team captain.
- Voted in for homecoming court. Placed 2nd in a vote for court king.

English International School of Prague

Elementary and Middle School

Prague, CZ

2007 – 2016

- Participated in two annual Worlds Scholars Cup competitions, focused on debate. Placed 1st in the Czech Republic debate round, winning a cup.
- Founded and participated in the Calculus Club for two years.
- Competed in cricket and practiced for the Duke of Edinburgh award.

EXPERIENCE

Atar Energy Solutions [thinktank]

Co-Founder

Aug 2024 – Present

Alexandria, VA

- Worked with Dr. Siavash Gholamhosseinpour to design, prototype, and build a novel FDS, Insulation Resistance and HiPOT Integrated Tester. It costs and weighs $\frac{1}{3}$ of the closest competition in a \$700M market
- Consulted with a handful of IP-law firms and negotiated deals with several potential private investors. Created the plan for non-provisional patent acquisition and legal entity creation
- Edit a 120-page technical datasheet and a 70-page professional plan

Curious Cardinals

Student Mentor

March 2025 – Present

Alexandria, VA

- Provide 1:1 project-based mentoring to 6 middle and high school students per 10-week cycle.
- Guide students through hands-on builds like a musical analysis website, custom-gear mechanical clock, and Raspberry Pi arcade console.
- Tailor STEM curriculum to individual learning styles, prototyping skills, and professional documentation.

Magellan Tutoring

Tutor

Jan 2025 – Present

Alexandria, VA

- Tutor high school and early college students in calculus, trig and precalc, physics, and computer science.
- Work with 8-10 students weekly on a 10-week schedule. Focus on working with students who have ADHD.

Campus Enterprises - Duke

Math and Physics Tutor

June 2022 – March 2024

Durham, NC

- Facilitated successful operations for Duke's GoBringIt food delivery, assisting with scheduling, logistics, and technical troubleshooting.

- On busy days, would manage and collect over two thousand dollars.

Duke General Robotics Lab

Aug 2022 – May 2024

Individual Research

Durham, NC

- Worked with Dr. Boyuan Chen to build and develop a versatile and accurate particle parallel-processed wave propagation simulation. This could emulate any material and lattice, 3D shape, and input forces. Used to generate artificial data to train a custom A.I. that can "see with sound"
- Created a novel Chladni plate comparison test to connect simulated data to real-world vibration data. See website portfolio for the white paper and results
- Research approved by Duke Engineering Board for *Graduation with Distinction award*

Starbucks

Feb 2021 – Aug 2022

Barista

Washington DC

- Managed high-pressure operations during rush hours (100+ customers/hour), coordinating 4-person teams and maintaining quality standards.
- Trained 3 new employees on POS systems and beverage protocols.
- Brought many smiles.

FPrin LLC

Oct 2020 – Feb 2021

Lab Technician

Mountain View, CA

- Designed, built, and operated case-specific test devices to effectively characterize a recently patented German compressed gas-powered insulin pen. After that, stress tested thousands of components and maintained the shop/lab
- Designed and manufactured a fully functional self-balancing remote-controlled PID controlled robot named Balanciaga for as an internal project

Null Designs LLC [startup]

March 2020 – Aug 2020

Head of Electrical Engineering

Los Altos, CA

- Explored tactile and low-budget computer keyboard systems for a start-up that was founded right after graduation with investment from a friend.
- Designed the underlying PCB using the ATMEGA-32 chip firmware as a base in EaglePCB.
- Tested numerous keys for ease-of-use and compatibility. Integrated the PCB and housing to manufacture a one-off custom keyboard for customers.

PROJECTS

Handheld Information Hub Device

April 2025 – Present

Alarm System, Air Measuring System, Mini Games

- Designed and built a compact device using the Raspberry Pi Pico to measure and display temperature, humidity, pressure, and sound. Integrated with a round LED screen and a aesthetic form-factor. Battery life of four days on one charge.

- Developed a real-time graphics and UI library using python; implemented sensor-driven animations and user input interrupt button control.
- Optimized hardware layout and code performance for low memory micro-controllers, including I2C/SPI protocol management, power-efficient design, and custom PCB/shell prototyping in Onshape with FDM 3D printing.

Accurate GPU Based N-Body Simulation Study

Aug 2024– Dec 2024

Personal Project, Part of AI Training Data

- Developed a GPU-accelerated N-body simulator in Python using the Taichi framework, achieving real-time performance with over 400 interacting celestial bodies at 30+ FPS.
- Implemented and compared multiple numerical integrators—including Euler, RK4, RK5, and Adams-Bashforth—to balance speed and orbital accuracy.
- Integrated ephemeris data from JPL DE-430 using the Astropy library to simulate accurate solar system dynamics in Cartesian coordinates.
- Designed and tested a custom GUI to visualize 3D system behavior and track orbital divergence over time.
- Evaluated simulation fidelity using local and global error metrics in position and energy, validating integrator performance across time steps.

High-ISP Electro-resistive Thruster

Aug 2023 – May 2024

Senior Design Capstone, Group of Six

- Designed, modeled, and tested a novel Electric Resistive Thermal Propulsion (ERTP) engine intended for long-duration space missions, meeting client requirements aligned with future NASA research grants.
- Developed a full thermodynamic and fluid model in MATLAB to predict thrust, ISP, and burn time; validated results through physical testing and experimentation.
- Constructed a functioning prototype using stainless steel tubing, nichrome heating coils, and a custom-built thrust stand capable of measuring sub-Newton forces.
- Achieved a maximum thrust of 1.36 N and ISP of 162.3 s in prototype tests, confirming strong correlation with theoretical predictions.
- Integrated safety-focused design practices using thermal insulation, regulated power, and structural fail-safes; maintained \$2000 project budget.
- Delivered comprehensive technical documentation and experimental results in a proposal-ready format for NASA consideration.

Personal Portfolio Website

April 2023 – Present

Live and Responsive JavaScript, Web Development, www.danielghasemfar.com

- Designed and deployed a responsive, multi-page portfolio website using HTML, CSS, JavaScript, and p5.js, hosted via GitHub Pages on a custom domain name.
- Integrated dynamic and animated custom art-inspired backgrounds using real-time JavaScript rendering with efficient memory handling and user-controlled themes.
- Optimized for both desktop and mobile devices, resolving scroll-related resets and file path compatibility across platforms.

Bipedal Walking and Dancing Robot

Jan 2023 – May 2023

Robotics Design Studio - Collaboration with a Partner

Duke University

- Designed and fabricated a 2-foot-tall, 8-DOF bipedal robot using LX-16A servos and modular 3D-printed

components, controlled via a Raspberry Pi 4 over Wi-Fi. Capabilities included walking, running, salsa dancing, headbanging, and performing the splits.

- Developed a Python-based motion control system with custom keyframe interpolation using the LX-16A library, enabling wireless execution of dynamic movement sequences.
- Resolved complex hardware integration issues such as solder joint failures, mechanical instability, a broken hip, and iterative CAD redesigns, resulting in a fully functional prototype with an Iron Man-inspired finish.

GPU-Accelerated Wave Propagation Simulation

Aug 2022 – May 2024

Duke General Robotics Lab

Durham, NC

- Developed a novel Taichi/PyImpact framework for parallel-processed wave simulation, modeling arbitrary 3D materials/lattices with accurate force propagation. System generated synthetic datasets to train AI models for acoustic spatial perception ("seeing with sound")
- Created groundbreaking Chladni plate validation method, quantitatively matching simulated harmonic modes (92% accuracy) with experimental data through spring-mass-damper lattice optimization. Results published in departmental white paper
- Recognized with Duke's *Graduation with Distinction* for advancing physics-based simulation methods, including GPU-accelerated PDE solving and PyBullet collision integration

UR5e Robot Arm Minigolf and ROS

Oct 2022 – Dec 2022

Intro to Robotics Final Project

- Engineered a system for autonomous golf putting by integrating a 3D-printed golf club with a UR5e robotic arm, enabling accurate trajectory execution using Cartesian linear path planning.
- Designed CAD of the golf course, Python-based robot motion control, and simulation using ROS and Gazebo.
- Used simple color-based video segmentation to identify ball starting position via laptop camera feed.
- Configured and deployed ROS launch files to the physical UR5e arm.

Magnetic Levitation and Transfer Functions

Oct 2022

Control Theory Lab [Group of Four]

- Designed and implemented a closed-loop PID controller in LabVIEW to achieve stable levitation of a metallic ball using an electromagnetic coil.
- Derived and linearized governing equations for magnetic force and ball dynamics; constructed a transfer function model to inform PID tuning strategy.
- Tuned controller gains iteratively using Ziegler–Nichols with empirical testing.
- Validated controller performance across diverse inputs, achieving minimal overshoot and near-zero steady-state error.

PID controlled RC Self-balancing Two Wheeled Robot

Dec 2020 – Feb 2021

Internal White Paper for Fprin [LLC]

- Designed and prototyped a two-wheeled self-balancing robot using a reverse pendulum model and real-time PID control for dynamic stability.
- Developed onboard control firmware in Arduino IDE, incorporating closed-loop feedback from IMU sensors to adjust wheel torque.
- Integrated a wireless control system enabling directional input for forward/backward motion and turning, while maintaining balance when idle.

- Created mechanical CAD in SolidWorks and designed an Eagle-based PCB layout for motor and sensor integration; sourced components with a budget constraint under \$500.

Thrust Vector Controlled Model Rocket

July 2019 – Feb 2020

Funded Advanced Scientific Investigation Class

Mountain View, CA

- Designed and tested a dual-axis thrust vector control (TVC) system for a solid-fuel model rocket to improve dynamic stability during ascent.
- Developed a working TVC mount with an Arduino Nano, MPU6050 IMU, and high-speed servos, achieving directional control via PID algorithms optimized using latency testing and simulation.
- Simulated deviation correction with a custom Python PID controller and verified improvements using recorded latency and angular deflection data.
- 3D-modeled and iteratively tested aerodynamic fin and nose cone designs to augment TVC stability and reduce oscillation post-thrust.

Using CNN's to Detect Forest Fires in Video Feed

XX-YY

Funded Advanced Scientific Investigation Class

Mountain View, CA

- Developed a convolutional neural network in TensorFlow 2.0 to detect forest fires in drone and CCTV video feeds, achieving 94% classification accuracy on a 1,000-image test set.
- Created a custom training dataset of 9,000 images by programmatically generating synthetic samples through a collage-based image compositing algorithm using fire, smoke, and landscape elements.
- Tested additional model variants using infrared video data collected from a FLIR thermal camera, demonstrating proof of concept for multi-spectral fire detection.
- Explained convolutional filtering, kernel operations, and backpropagation mechanics in a science fair presentation. Awarded 2nd Place in Computer Science and Natural Sciences at the 2019 California Synopsys Science Fair Symposium. Received a medal from the Mountain View mayor for recognition of research at a young age.

EXTRACURRICULARS

Clubs & Activities & Hobbies

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| – High School XC Mens Captain | Physics SATs | – 2x Synopsys Science Fair |
| – 3x XC States Championships | – 6+ years Calculus/Trig Tutor | Finalist [CS] |
| – California Mayor Medal for A.I. forest fires study | – FRC Robotics Quarterfinals | – Los Altos Hackathon III organization |
| – 35 ACT Score, 800 Math and | – Duke 24-hour Design Sprint Winner | |